

# Recommendations — gist-sdh-multifocal-resected-m1-k4n8

*Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references*

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# Recommendations — gist-sdh-multifocal-resected-m1-k4n8

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## Profile snapshot

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- **Primary site:** stomach
- **Histology:** gastrointestinal stromal tumor (GIST), SDH-deficient subtype (KIT/PDGFRA wild-type)
- **Stage:** ypT3(m)N0M1 — multifocal gastric GIST with a single M1 deposit in the ligament of Treitz; R0 distal gastrectomy + Roux-en-Y; 0/7 nodes; largest mass 7.5 cm; M1 deposit completely resected
- **ECOG:** 1
- **Age band:** 40-49
- **Sex:** unknown
- **Biomarkers:** SDHA L306P mutated (VAF 44%; germline-vs-somatic not yet established) (ngs\_pending); SDHA Y408N mutated (VAF 41%, somatic); SDHA E350fs mutated (ctDNA) (ngs\_pending); PIK3CA R93W mutated (ctDNA) (ngs\_pending); MAP2K1 P124S mutated (ctDNA) (ngs\_pending); POLE R1679C mutated (germline); CD117 (KIT) protein positive; DOG1 protein positive; Desmin negative; SDHB protein intact; KIT wild-type; PDGFRA wild-type; TMB <1 mut/Mb; MSI status MSS; NTRK no fusion detected; RET no actionable alteration; BRAF wild-type
- **Efficacy/toxicity weight:** 0.65

*8 ranked therapeutic options across 2 targetable features. Biomarker workup steps live in the standalone Target validation paths report.*

## Sdh Deficient Gist interventions

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### 1. Signatera tumor-informed ctDNA MRD assay as recurrence-surveillance adjunct to imaging (informative, not therapy-triggering)

**Likelihood of effect:** Lead-time signal ahead of imaging — informative for recurrence monitoring; weaker prospective GIST validation than colorectal/lung MRD data.

**Toxicity burden:** Low (none — venipuncture; FFPE block re-use for baseline build)

**Counter-productive MoA:Low** (False-positive ctDNA without imaging correlate could drive premature TKI initiation; imaging-gating guardrail is the mitigation.)

**Overall:**Adjunct to imaging surveillance with a clear guardrail — a positive Signatera prompts shorter-interval imaging, not systemic therapy.

**Key references:** PMID 34754095 · PMID 34380780

### 2. Sunitinib 50 mg PO daily 4-on/2-off (or 37.5 mg continuous) — first systemic line at measurable recurrence; NOT adjuvant

**Likelihood of effect:** Moderate at first measurable recurrence — Boikos 2016 dSDH-GIST 18% ORR (n=38 retrospective, no CI); Demetri 2006 registration-grade 2L GIST (TTP HR 0.33).

**Toxicity burden:** Moderate (HFSR, hypothyroidism, fatigue, hypertension — well-characterized chronic class AEs; published management algorithms apply)

**Counter-productive MoA:Low** (Class-effect VEGFR signaling withdrawal on chronic dosing; no mechanism-level dissent for THIS subset in the board's deliberation.)

**Overall:Best-characterized 2L TKI for KIT/PDGFR-WT GIST; contingency-only at measurable recurrence, with the trial-route options as a co-equal conversation rather than a downstream fallback.**

**Key references:** PMID 17046465 · PMID 27011036 · PMID 40045030

### 3. Regorafenib 160 mg PO 3-on/1-off — 3L contingency at sunitinib progression

**Likelihood of effect:** Low-to-moderate in dSDH-GIST — GRID PFS HR 0.27 is registration-grade in KIT-mutant 3L; dSDH-subset activity is by extrapolation, no published subset CI.

**Toxicity burden:** Moderate (G3+ hypertension 23%, G3+ HFSR 20%, G3+ diarrhea 5%; dose modifications are the rule)

**Counter-productive MoA:Low** (Advocate dissent was preference-fit on sequencing against the trial-route, not on the mechanism.)

**Overall:Registration-grade 3L anchor with weak dSDH-GIST subset evidence; sequencing against the trial-route at 3L is the live preference question.**

**Key references:** PMID 23177515 · PMID 40045030

### 4. Adjuvant imatinib — NOT RECOMMENDED in dSDH-GIST (logged on the record so the no-imatinib decision is documented and not re-litigated)

**Likelihood of effect:** None in dSDH-GIST — Boikos 2016 1/49 (2%) PR on imatinib; adjuvant trials structurally excluded this subset.

**Toxicity burden:** High (3-year exposure: G2+ edema ~30%, fatigue ~50%, hepatic / cardiac monitoring burden) measured against a 2% PR rate — not a defensible tradeoff.

**Counter-productive MoA:High** (Both vetoes stood on mechanism + toxicity: imatinib insensitivity in dSDH-GIST plus chronic AE burden at the 2% PR floor; PERSIST-5 confirmed imatinib-insensitive genotypes recur regardless of treatment.)

**Overall:Logged as NOT RECOMMENDED so the reflex 'GIST equals imatinib' reach is documented and rejected on the record — every persona vetoed in concept.**

**Key references:** PMID 27011036 · PMID 30383140 · PMID 19303137 · PMID 22453568 · PMID 32469385

### Evidence in detail

##### Sunitinib 50 mg PO daily 4-on/2-off (or 37.5 mg continuous) — first systemic line at measurable recurrence; NOT adjuvant

**\*\*Demetri GD et al. *Lancet* 2006\*\***

- **Disease context:**advanced/metastatic GIST after imatinib failure or intolerance (2L+) — 312 patients (207 sunitinib, 105 placebo) with imatinib-resistant or -intolerant GIST. No SDH-status stratification; the cohort was overwhelmingly KIT-mutant by historical molecular epidemiology, with dSDH-GIST under-represented.
- **n:** 312
- **Toxicities:**fatigue (grade any); diarrhea (grade any); skin discoloration (grade any); nausea (grade any)
- **Efficacy:**TTP 27.3 weeks; TTP 6.4 weeks; HR\_PFS 0.33

**\*\*Boikos SA et al. *JAMA Oncology* 2016\*\***

- **Disease context:**KIT/PDGFRA wild-type GIST (NIH Pediatric & Wild-Type GIST Clinic cohort; majority SDH-deficient) (2L+) — 95 patients with adequate tumor specimens from the NIH wt-GIST clinic; median age 23; 70% female; molecular subtypes: SDH-mutant (n=63), SDH-epimutant (n=21), SDH-competent (n=11). The cohort skews young and pediatric-onset, with gastric primaries dominating the SDH-deficient strata.
- **n:** 95
- **Toxicities:** Toxicity not the focus of this cohort report; sunitinib and imatinib safety as expected for label dosing.
- **Efficacy:**ORR 2%; ORR 18%

##### Regorafenib 160 mg PO 3-on/1-off — 3L contingency at sunitinib progression

**\*\*Demetri GD et al. *Lancet* 2013\*\***

- **Disease context:**advanced GIST after imatinib + sunitinib failure (3L+) — 199 patients (133 regorafenib, 66 placebo) with progression after both imatinib and sunitinib. No prospective SDH-status stratification; KIT-mutant disease dominated the population.
- **n:** 199
- **Toxicities:**hypertension (grade  $\geq 3$ ): 23% (n=132); hand-foot skin reaction (grade  $\geq 3$ ): 20% (n=132); diarrhea (grade  $\geq 3$ ): 5% (n=132)
- **Efficacy:**PFS 4.8 months; PFS 0.9 months; HR\_PFS 0.27 (95% CI 0.19-0.39)

##### Adjuvant imatinib — NOT RECOMMENDED in dSDH-GIST (logged on the record so the no-imatinib decision is documented and not re-litigated)

**\*\*Boikos SA et al. *JAMA Oncology* 2016\*\***

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- **n:** 95
- **Toxicities:** Toxicity not the focus of this cohort report; sunitinib and imatinib safety as expected for label dosing.
- **Efficacy:**ORR 2%; ORR 18%

**\*\*Raut CP et al. *JAMA Oncology* 2018\*\***

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

**\*\*DeMatteo RP et al. *Lancet* 2009\*\***

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

**\*\*Joensuu H et al. *JAMA* 2012\*\***

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

**\*\*Joensuu H et al. *JAMA Oncology* 2020\*\***

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

## Sdh Confirmed interventions

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### **1. Belzutifan 120 mg PO daily on LITESPARK-015 wt-GIST cohort (NCT04924075) — contingency at first measurable recurrence, gated on dSDH phenotype confirmation \*(conditional on sdh\_confirmed positive)\***

**Likelihood of effect:** Moderate if dSDH phenotype confirms — PPGL adjacency (ORR 26%, mPFS 22 mo) does not quantitatively transport to GIST; cleanest mechanism case but no GIST efficacy readout yet.

**Toxicity burden:** Moderate (anemia and hypoxia dose-limiting in 25-30% on broader dataset; stacked anemia management problem in Roux-en-Y; ~5-year post-marketing surveillance)

**Counter-productive MoA:Low** (Cross-tumor mechanism transport from PPGL to GIST is the gating uncertainty — modality-level risk, not patient-AE risk.)

**Overall:**Cleanest mechanism case for dSDH-GIST in the dossier — 14 US sites recruiting; cross-tumor transport from PPGL is the live question and Roux-en-Y anemia management is the patient-specific complication.

**Key references:** NCT04924075 · PMID 15652751 · PMID 35324464 · PMID 27595394

### **2. Temozolomide on NCT05661643 (Asan, 200 mg/m<sup>2</sup> d1-5 q28d) or NCT03556384 (UCSD, 85 mg/m<sup>2</sup> d1-21 q28d, watch-for-readout); off-label generic TMZ with MGMT methylation as enrichment lever — contingency at first measurable recurrence \*(conditional on sdh\_confirmed positive)\***

**Likelihood of effect:** Low-to-moderate — Yebra 2022 2/5 PR (95% CI ~5-85%) is mechanism-coherent but uninformative at n=5; Ligon 2023 guadecitabine 0/9 is the cautionary precedent for mechanism-vs-clinic disconnect in this subset.

**Toxicity burden:** Moderate (myelosuppression, fatigue, nausea well-characterized; MDS/AML long-tail ~1% at cumulative exposure >6-12 months is the chronic signal)

**Counter-productive MoA:** **Moderate** (Critic dissented on the Yebra n=5 point-estimate framing; Ligon 2023 (guadecitabine 0/9) is the mechanism-coherent precedent where the hypomethylator/alkylator class disconnect already played out in this subset.)

**Overall:** Mechanism-coherent contingency with mature parent-drug safety, but n=5 efficacy + the Ligon 2023 negative precedent argue for trial-route, MGMT-enriched, with a pre-specified cumulative-exposure stopping rule.

**Key references:** PMID 34426440 · PMID 38132569 · PMID 36198483 · NCT05661643 · NCT03556384

### **3. Pemigatinib on PEMIGIST (NCT07434843, DFCI single-site, Suzanne George) — contingency at first measurable recurrence \*(conditional on sdh\_confirmed positive)\***

**Likelihood of effect:** Low-to-moderate — rogaratinib NCT04595747 41.7% ORR at n=24 (95% CI ~22-63%) is the cited bridge; pemigatinib NCT07434843 itself is n=24 single-arm phase 2 with no readout yet.

**Toxicity burden:** Moderate (FGFR-class: serous retinal detachment ~25%, hyperphosphatemia ~60% all-grade, nail/mucosal AEs; monitoring-intensive class)

**Counter-productive MoA:** **Moderate** (Critic and advocate dissented on evidence-pyramid narrowness and preference-fit; FGFR-class toxicity-as-monitoring-burden is the mechanism-adjacent concern.)

**Overall:** Freshest mechanism story in the dossier with the strongest preclinical-to-clinic signal so far, but the evidence pyramid is shallow and two personas dissented on rank-1 framing.

**Key references:** NCT07434843 · doi:10.1158/1538-7445.AM2025-CT215

### **4. Olaparib 300 mg PO BID + temozolomide — late-line / post-trial-exhaustion contingency, trial-route only with hematology co-management (NOT off-label cash-pay default) \*(conditional on sdh\_confirmed positive)\***

**Likelihood of effect:** Low for population-level effect — Singh 2023 is n=1; Sulkowski mechanism work is robust at chromatin level but no replicated clinical readout in dSDH-GIST.

**Toxicity burden:** High (overlapping PARP + alkylator cytopenias; cumulative MDS/AML risk on sustained exposure; no published combination safety dataset in dSDH-GIST)

**Counter-productive MoA:** **High** (Conservative veto stood on chronic-toxicity-as-mechanism: PARP + alkylator MDS/AML compounding in sustained-response scenarios is the mechanism-level risk to durable use.)

**Overall:** Mechanism-strong, evidence-thin late-line contingency — acceptable only in bounded trial-route or hematology-comanaged framing; off-label cash-pay default is not signable.

**Key references:** PMID 30013182 · PMID 32494005 · PMID 36151992

## Evidence in detail

##### Temozolomide on NCT05661643 (Asan, 200 mg/m<sup>2</sup> d1-5 q28d) or NCT03556384 (UCSD, 85 mg/m<sup>2</sup> d1-21 q28d, watch-for-readout); off-label generic TMZ with MGMT methylation as enrichment lever — contingency at first measurable recurrence

**\*\*Yebra M et al. *Clinical Cancer Research* 2022\*\***

- **Disease context:**advanced **SDH-mutant GIST (SDHA/SDHB/SDHC/SDHD mutation, biallelic in tumor)** (2L+) — 5 SDH-mutant GIST patients (1 SDHA, 2 SDHB, 1 SDHC, 1 SDHD); median age 29.7; 2F/3M; median 1 prior TKI line (range 0-2). Sicklick / UCSD case series paired with patient-derived xenograft modeling.
- **n:** 5
- **Toxicities:** Paper does not publish a per-AE safety table for the clinical cohort; the translational focus is on the PDX-driven mechanism rationale. Standard TMZ low-dose continuous schedule toxicity profile (cytopenias, fatigue, nausea) presumed.
- **Efficacy:**ORR 40%; **DCR** 100%; **OS** 1.9 years

**\*\*Giger OT et al. *Journal of Clinical Pathology* 2022\*\***

- **Disease context:**SDH-deficient and **KIT/PDGFRA-WT GIST** (any) — 65 GIST tumors profiled: 47 wt-GIST (33 SDH-deficient), 21 tyrosine-kinase-mutant. Single-institution Cambridge cohort.
- **n:** 65
- **Toxicities:** —
- **Efficacy:**biomarker 24%

##### Olaparib 300 mg PO BID + temozolomide — late-line / post-trial-exhaustion contingency, trial-route only with hematology co-management (NOT off-label cash-pay default)

**\*\*Singh C et al. *Pediatric Blood & Cancer* 2023\*\***

- **Disease context:**metastatic, multiply-relapsed **SDH-deficient GIST + paraganglioma (single patient)** (5L+) — Single pediatric/young-adult patient with SDHB-mutant GIST plus paraganglioma; multiply relapsed after prior TKI lines. Yale single-institution case report.
- **n:** 1
- **Toxicities:** Case-report; per-AE detail not in the published letter.
- **Efficacy:**other clinical response